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(ii) Volume stress

This happens when a body is acted by forces everywhere on the surface such that the force at any point is normal to the surface and the magnitude of the force on a small surface area is proportional to the area. For instance, when a solid is immersed in a fluid, the pressure at the location of the solid is P, the force on any area ΔA is

$$F = P\,\Delta A$$

Where, F is perpendicular to the area. Thus, force per unit area is called volume stress.

$$\sigma_v = \frac{F}{A}$$

which is the same as the pressure.

(b) Strain:

Strain measures how much an object is stretched or deformed when a force is applied. Strain deals with the fractional change in the size of the object, in other words, strain measures the degree of deformation. As an example, in one dimension, consider a rod of length l when it stretches to a new length Δl then

$$\text{Strain, } \varepsilon = \frac{\text{change in size}}{\text{original size}} = \frac{\Delta l}{l} \qquad (7.2)$$

ε is a dimensionless quantity and has no unit. Strain is classified into three types.

(1) Longitudinal strain

When a rod of length l is pulled by equal and opposite forces, the longitudinal strain is defined as

$$\varepsilon_l = \frac{\text{increase in length of the rod}}{\text{original or natural length of the rod}} = \frac{\Delta l}{l} \qquad (7.3)$$

Longitudinal strain can be classified into two types

(i) Tensile strain: If the length is increased from its natural length then it is known as tensile strain.

(ii) Compressive strain: If the length is decreased from its natural length then it is known as compressive strain.

(2) Shearing strain

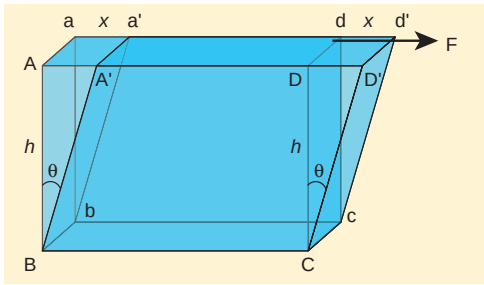


Figure 7.6 Shearing strain

Consider a cuboid as shown in Figure 7.6. Let us assume that the body remains in translational and rotational equilibrium. Let us apply the tangential force F along AD such that the cuboid deforms as shown in Figure 7.6. Hence, shearing strain or shear is (ε_s)

$$\varepsilon_s = \frac{AA'}{BA} = \frac{x}{h} = \tan\theta \qquad (7.4)$$

For small angle, $\tan\theta \approx \theta$

Therefore, shearing strain or shear,

$$\varepsilon_s = \frac{x}{h} = \theta = \text{Angle of shear}$$

(3) Volume strain

If the body is subjected to a volume stress, the volume will change. Let V be the original volume of the body before stress and $V + \Delta V$ be the change in volume due to stress. The volume strain which measures the fractional change in volume is

$$\text{Volume strain, } \varepsilon_v = \frac{\Delta V}{V} \qquad (7.5)$$

Elastic Limit

The maximum stress within which the body regains its original size and shape after the